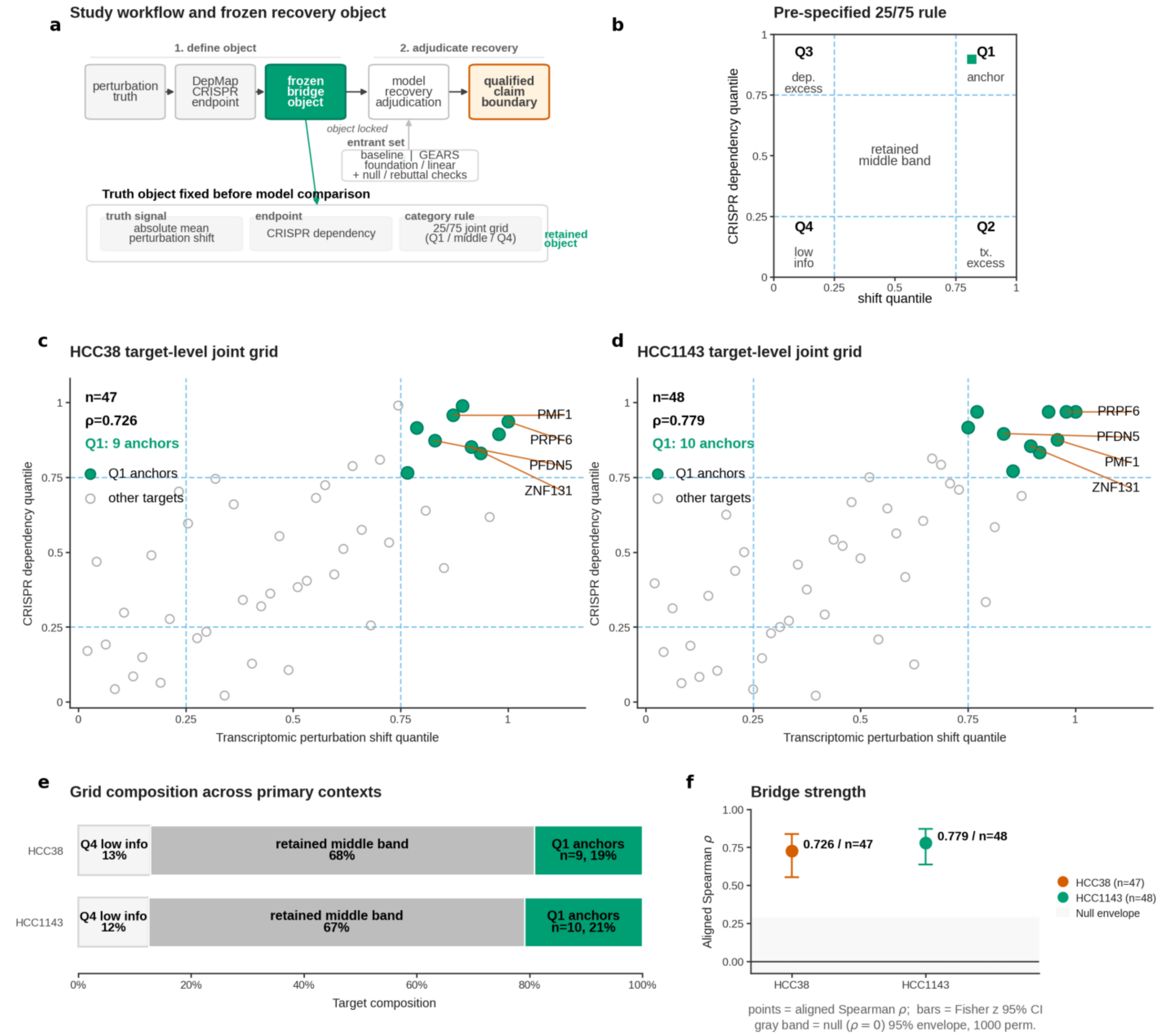


Fig. 1 | Truth object definition. a, Study workflow and frozen recovery object. b, Pre-specified 25/75 category rule. c, HCC38 target-level joint grid. d, HCC1143 target-level joint grid. e, Grid composition across the two primary contexts. f, Bridge strength summary.

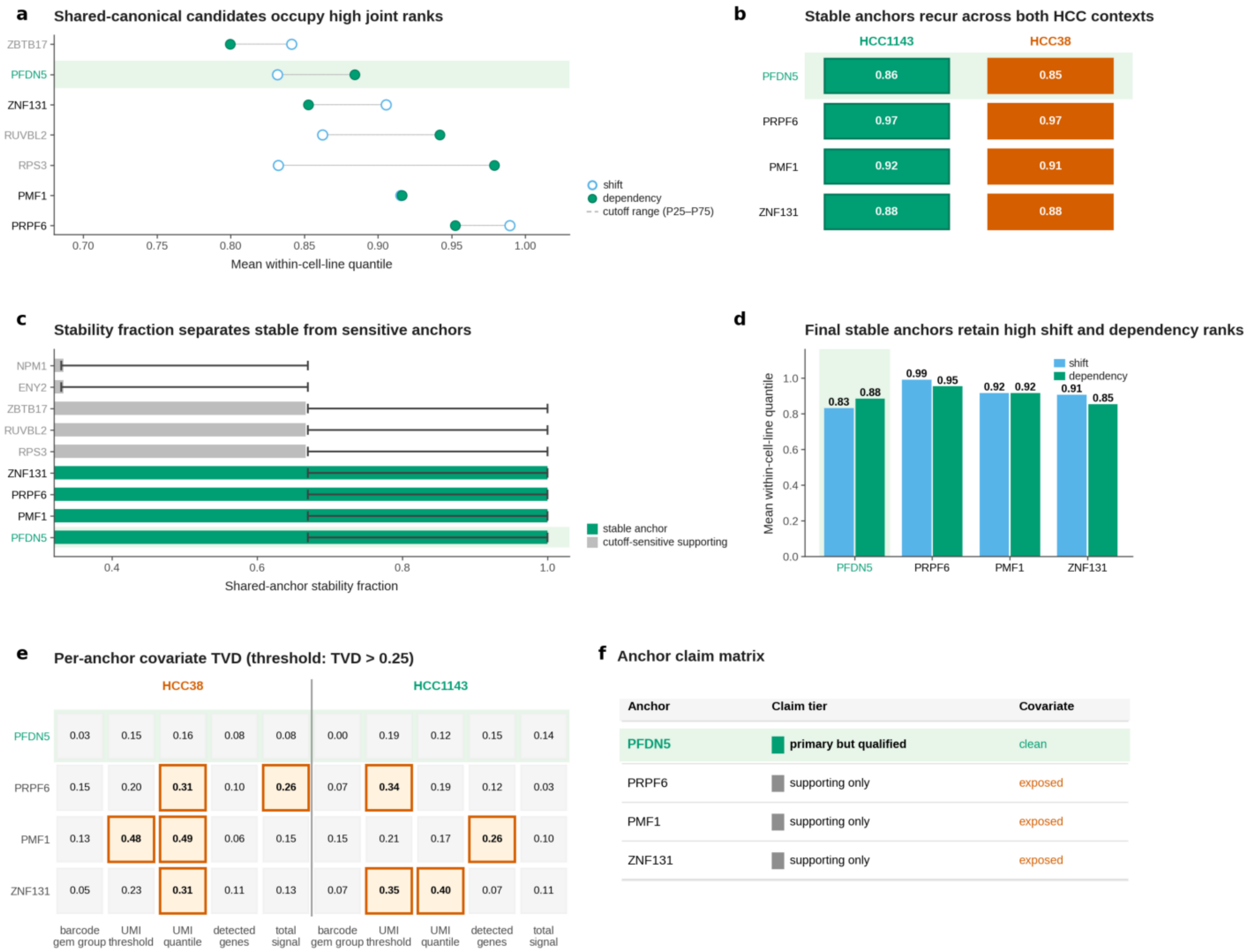
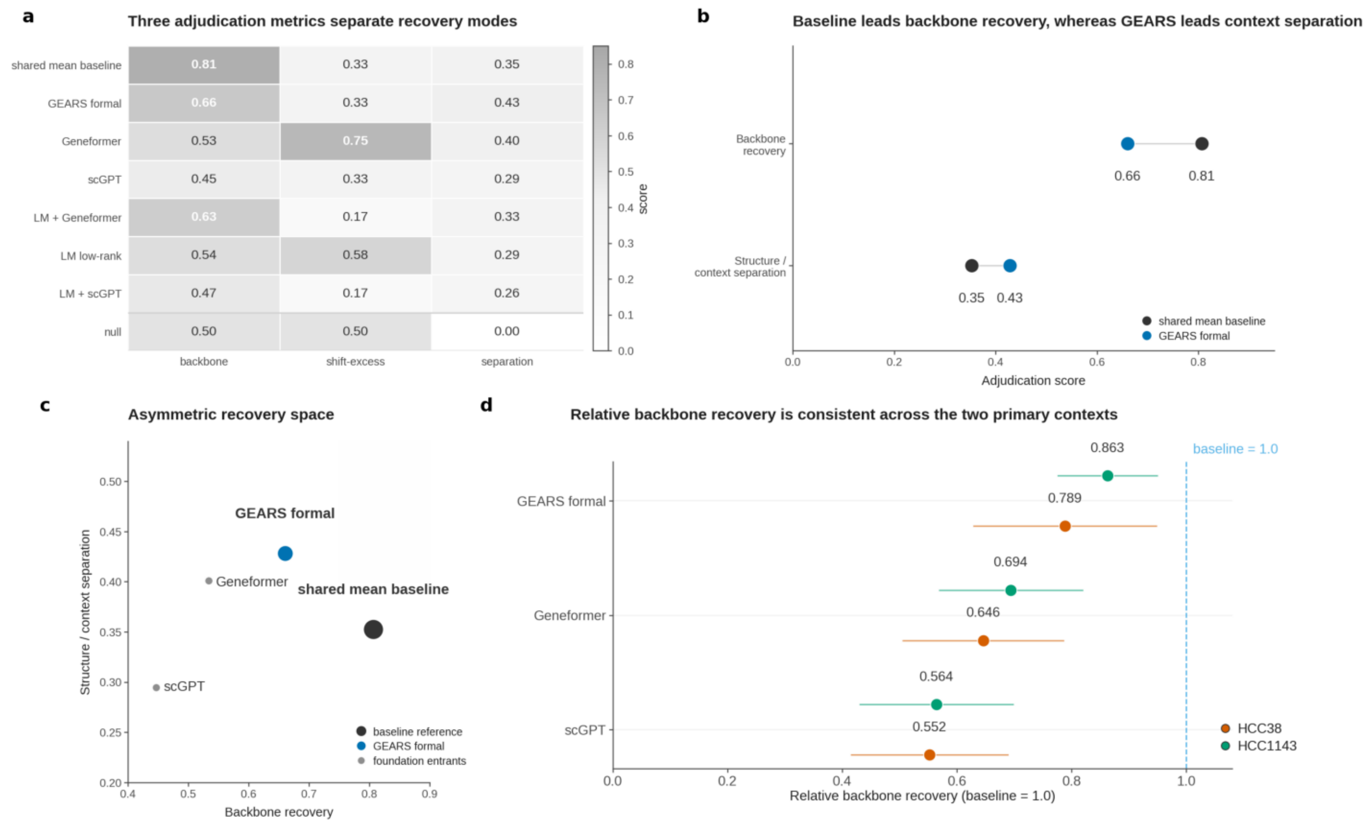


Fig. 2 | Anchor tiering. a, Shared-canonical anchor ranking. b, Low-ink recurrence matrix across the two primary HCC contexts (HCC1143 and HCC38). c, Stability fraction across stable and cutoff-sensitive shared-canonical objects. d, Final stable anchors retain high shift and dependency ranks across HCC38 and HCC1143. e, Per-anchor covariate Total Variation Distance (TVD) matrix across five covariate axes. f, Compact anchor claim matrix (conclusion).



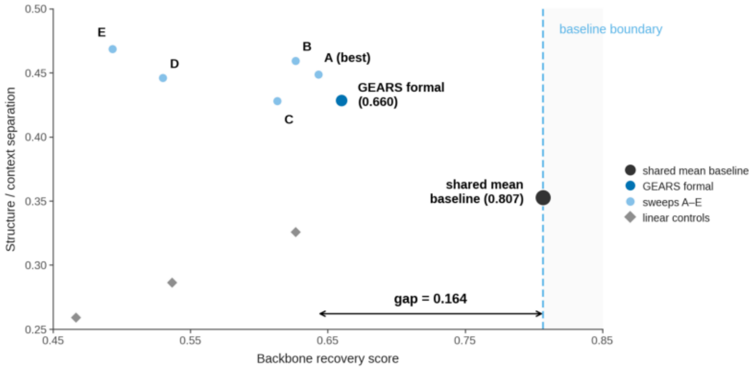
	backbone	shift-excess	separation
shared mean baseline	0.81	0.33	0.35
GEARS formal	0.66	0.33	0.43
Geneformer	0.53	0.75	0.40
scGPT	0.45	0.33	0.29
LM + Geneformer	0.63	0.17	0.33
LM low-rank	0.54	0.58	0.29
LM + scGPT	0.47	0.17	0.26
null	0.50	0.50	0.00

Fig. 3 | Model adjudication. a, Three adjudication metrics separate recovery modes across entrants. b, Baseline leads backbone recovery, whereas GEARS leads context separation. c, Entrants occupy an asymmetric recovery space.

a Pre-specified finite-budget GEARS candidates

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate

b Prespecified rebuttal candidates do not close the backbone gap



c No tested rebuttal candidate closes the backbone gap to the shared-mean baseline

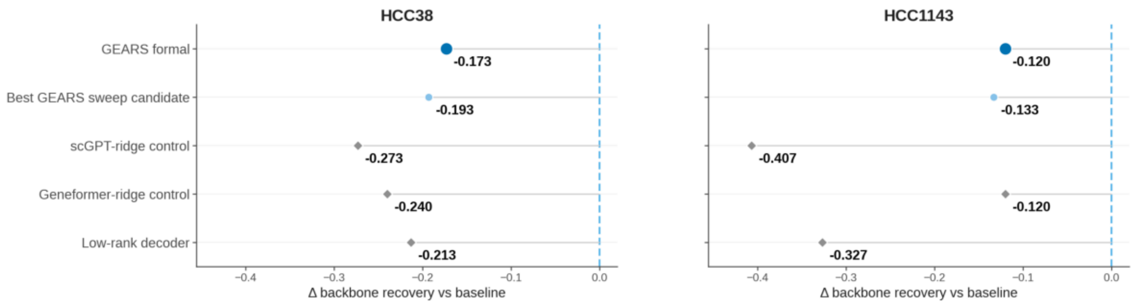


Fig. 4 | Finite-budget rebuttal tests. a, Pre-specified finite-budget GEARS candidates. b, Pre-specified rebuttal recovery space. c, Per-context residual backbone gap.

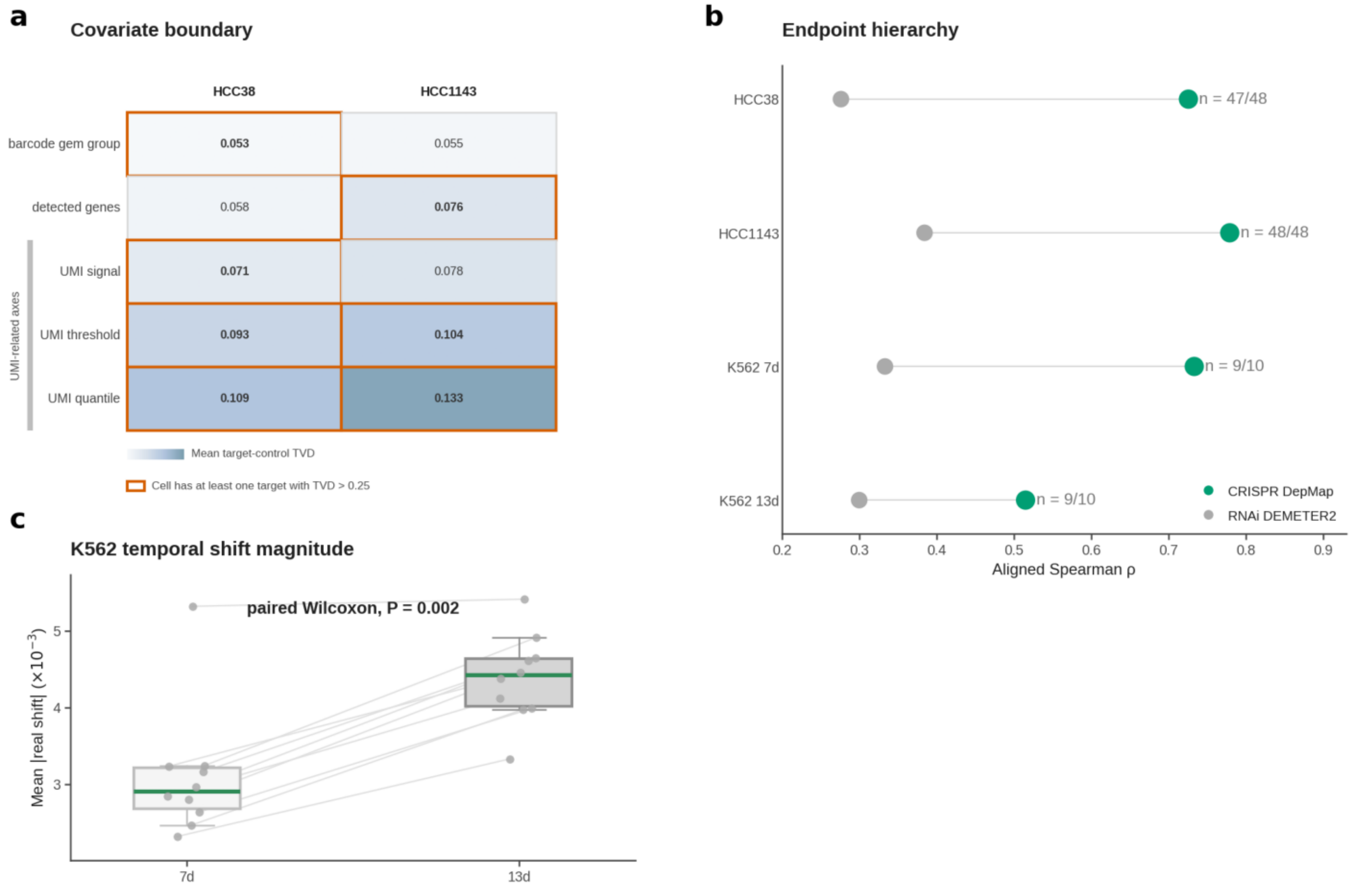


Fig. 5 | Boundary governance. a, Covariate boundary in HCC38 and HCC1143 (Framework element 7). b, Endpoint hierarchy across all four bridge contexts (Framework element 6). c, K562 temporal stratification: stronger rank alignment at 7d despite smaller perturbation magnitude. d, A0/A1/B evidence-tier summary for K562 temporal evidence (Framework element 6).